

SIVASANKARAN E

B.Tech — Artificial Intelligence & Data Science

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PROFESSIONAL SUMMARY

Motivated Artificial Intelligence and Data Science undergraduate with a strong foundation in Full-Stack Development, Python, JavaScript, SQL, and AI/ML. Skilled in developing responsive web applications and RESTful APIs using React.js, Node.js, Flask, and PostgreSQL/MySQL, with hands-on experience integrating Generative AI and LLM APIs into applications. Familiar with data processing, machine learning workflows, database design, authentication, and process automation. Passionate about building scalable, intelligent, and user-focused software solutions while continuously improving technical and problem-solving skills.

SKILLS

Languages: Python, SQL, JavaScript

Backend & Frameworks: Node.js, Express.js, Flask, REST APIs, React

Databases: PostgreSQL, MySQL, SQLite, Supabase

API & Integration: RESTful APIs, JSON, LLM API Integration (Groq, Gemini), API Authentication

Data & Automation: Pandas, NumPy, n8n, Playwright, APScheduler, ETL

Security & Backend Concepts: Token-Based Authentication, RBAC, Parameterized SQL, CRUD Operations, Database Design, Data Validation

Developer Tools: Git, GitHub, VS Code, Postman, Jupyter Notebook, Google Colab.

Additional: Machine Learning, Generative AI, Prompt Engineering, Process Automation.

PROFESSIONAL EXPERIENCE

Web Development Intern — Macx Technologies

Feb 2024 – Mar 2024

- Contributed to front-end development by building responsive web interfaces with HTML, CSS, JavaScript, and React.
- Gained hands-on exposure to real-world development workflows, including Git version control, code reviews, and REST API integration.

PROJECTS

Freewheel — Agency Operations & Resource Management Platform

Jul 2026 – Aug 2026

Node.js, Express.js, PostgreSQL, Supabase, REST APIs, pg-pool

- Developed a modular RESTful API backend using Node.js and Express.js for authentication, team management, task boards, and billing.
- Designed a normalized 7-table PostgreSQL schema with B-Tree indexes, foreign keys, JSONB, and database constraints.
- Implemented RBAC, token-based authentication, and parameterized SQL queries to ensure secure data access and prevent SQL injection.
- Built automated time tracking, project budget monitoring, and invoice generation with real-time data aggregation.
- Deployed the application on Render with Supabase, PostgreSQL connection pooling, PM2, and offline caching/failover support.

Learning Progress Tracker Dashboard

Aug 2025 – Nov 2025

Python, MySQL, Streamlit

- Built a personal learning management web app enabling users to log courses, set weekly goals, track completion status, and visualize learning velocity over time.
- Architected a normalized MySQL database with tables for topics, resources, progress logs, tags, and priority ratings; enforced referential integrity with foreign key constraints.
- Enabled dynamic filtering by domain, status, and priority with a Streamlit UI, allowing learners to surface high-priority pending items instantly.
- Added an analytics module computing weekly study hours, streak tracking, and completion-rate trends rendered as interactive Plotly charts.
- Applied secure credential handling with python-dotenv and structured the codebase using modular MVC-style separation for maintainability.

GitHub AI Repo Scout — Automated Project Discovery Bot

Jul 2026

n8n, GitHub REST API, Groq LLM API, Telegram Bot API, Docker

- Built an automated n8n workflow, self-hosted via Docker, that queries the GitHub REST API on a daily schedule to discover trending repositories tagged with AI-related topics.
- Designed a multi-stage data pipeline — splitting, cleaning, and language-filtering raw API results — to normalize repository metadata before scoring.
- Integrated the Groq LLM API to automatically score each repository's suitability as a final-year project idea (1–10) against a defined tech stack, with structured reasoning parsed from the model's JSON output.
- Built a relevance filter to surface only high-scoring project ideas and delivered real-time alerts to a Telegram bot with project links, fit scores, and reasoning.
- Managed end-to-end workflow orchestration and container deployment, monitoring execution logs and node-level outputs for reliability.

WattWise — Smart Energy Optimization Platform

Feb 2024 – Mar 2024

Next.js, React, TailwindCSS, Google Genkit (Gemini AI)

- Built a full-stack AI-powered energy monitoring and optimization platform using Next.js 15 (App Router), React 18, and TailwindCSS, with a shadcn/ui and Radix UI component library for a responsive dashboard experience.
- Developed an interactive single-pane-of-glass energy dashboard integrating real-time consumption monitoring, CSV/JSON historical log uploads, Recharts visualizations, and goal-setting and tracking widgets.
- Engineered three Google Genkit AI flows powered by Gemini 2.0 Flash: an energy consumption predictor, a personalized energy-saving advisor, and a conversational energy assistant chatbot.
- Implemented TanStack Query (react-query) for efficient client-side state management and asynchronous data fetching across dashboard widgets.
- Integrated Firebase SDK for backend data persistence and configured Turbopack for optimized, fast production builds.

Internship Application Automation System

Jun 2026 – Jul 2026

Python, Playwright, SQLite, Streamlit, LLM APIs

- Built an end-to-end automation pipeline to scrape internship listings from Internshala, LinkedIn, and Unstop using Playwright and automatically fill application forms.
- Designed a custom matching engine to filter and rank internship listings based on user profile criteria.
- Integrated Groq Llama 3.1 70B with Google Gemini as a fallback to generate personalized cover letters using free-tier LLM APIs.
- Implemented a local-first architecture using SQLite, Streamlit, and APScheduler for data storage, monitoring, and scheduled automation.
- Added Gmail SMTP alerts and deployed the system as a free, self-hosted solution on local PC, Google Colab, and Oracle Cloud Free Tier.

EDUCATION

B.Tech — Artificial Intelligence & Data Science, IFET College of Engineering, Villupuram, Tamil Nadu
2023 – 2027

CGPA: 8.3 / 10.0

- Relevant coursework: Machine Learning, Data Structures & Algorithms, Database Management Systems, Artificial Intelligence, Big Data Analytics, Backend Development, AI Ethics & Societal Impact.

Higher Secondary (Class XII) — Computer Science, St. John's Higher Secondary School, Tamil Nadu
2022 – 2023

Percentage: 89%

CERTIFICATIONS

- Foundations of Prompt Engineering — AWS Training & Certification, Jul 2026
- Data Processing and Visualisation — NASSCOM FutureSkills Prime, Jul 2026
- Introduction to Data Engineering — IBM (Coursera), Mar 2026
- Claude Code in Action — Anthropic, Feb 2026
- Generative AI for All — Infosys Springboard, Jan 2026
- Data Analytics Professional Certificate (30-Day MasterClass) — NoviTech R&D, Feb–Apr 2025

ACHIEVEMENTS & ACTIVITIES

- Active member of the college AI & Data Science Club — conducted workshops on Python and data visualization for junior students.
- Prepared research-level study material on AI ethics covering fairness, transparency, accountability, and societal impacts across healthcare, finance, and social media.
- Designed and self-hosted workflow automation pipelines using n8n and Docker, integrating REST APIs and LLMs to automate research, scoring, and real-time alerting tasks.